Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

NAAC 'A' Grade Accredited

Department of Information Technology

UG Programme in IT NBA Accredited

MINI PROJECT SYNOPSIS

(Academic Year 2026-2027)

Project Group Id	B3-03
Title of the Project	**Shopping List Extractor:** An NLP-Based System for Extracting and Organizing Shopping Items from Natural Language Text
Project Domain	Artificial Intelligence – Natural Language Processing (NLP) / Machine Learning
Problem Statement	People often create shopping lists using natural language, such as *"I need two packets of biscuits, one litre of milk, three apples and some shampoo."* Such information may be written in messages, notes, emails, or other unstructured text. Manually identifying each shopping item, its quantity, and related information can be time-consuming and may lead to missing or incorrectly recorded items. The proposed system, **Shopping List Extractor**, is designed to automatically identify and extract shopping-related information from natural language text. The system uses Natural Language Processing techniques to understand the input and identify important entities such as **item names, quantities, units, and categories**. The extracted information is then organized into a structured shopping list. The system can process user-provided text and convert it into a clear and organized list. For example, a sentence such as *"Buy 2 kg rice, 1 litre milk and 5 apples"* can be converted into structured entries such as **Rice – 2 kg, Milk – 1 litre, Apples – 5**. The system aims to reduce manual effort and improve the accuracy and usability of shopping-list creation.
References	[1] R. Loughnane, J. Liu, Z. Chen, Z. Wang, J. Giroux, T. Du, B. Schroeder, and W. Sun, "Explicit Attribute Extraction in e-Commerce Search," *Proceedings of the Seventh Workshop on e-Commerce and NLP (ECNLP 7)*, pp. 125–135, 2024. [2] S. Arslan, "Application of BiLSTM-CRF Model with Different Embeddings for Product Name Extraction in Unstructured Turkish Text," *Neural Computing and Applications*, vol. 36, pp. 8371–8382, 2024. [3] K. Roy, P. Goyal, and M. Pandey, "Exploring Generative Frameworks for Product Attribute Value Extraction," *Expert Systems with Applications*, vol. 243, 2024.
Aim of the Project	The goal of the project is to develop an NLP-based Shopping List Extractor that automatically extracts shopping items and their relevant information from natural language text. The system will identify item names, quantities, units, and categories and organize them into a structured shopping list. It will provide a simple and efficient way for users to convert unstructured shopping-related text into an easily understandable list.

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